

1. Origin
2. SOP (Same-Origin Policy)
3. Cross-Origin Request
4. CORS (Cross Origin Resource Sharing)
5. Preflight Request
6. Practical

Origin:

Protocol + Domain + Port

Example:

http://localhost:3000
 https://loopkaka.com:2000
 http://google.com
 https://subdomain.google.com

Same Origin vs Different Origin

http://localhost:3000/test
 http://localhost:3000/abc

Same ORigin

http://localhost:3000
 http://localhost:2000

 http://loopkaka.com
 https://loopkaka.com

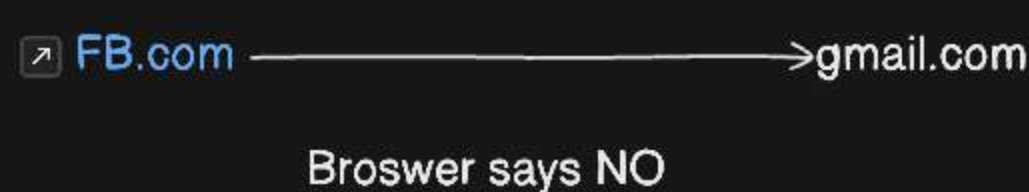
 https://subdomain.loopkaka.com
 https://loopkaka.com

 https://loopkaka.com
 https://google.com

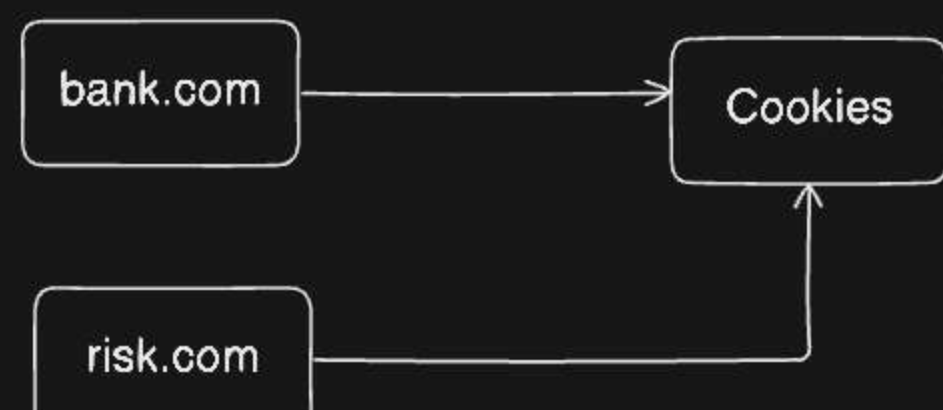
Different Origin

SOP (Same-Origin Policy)

You can access anything which is on same domain but can't access which are on other domain



- By default browser enable same origin policy
- It's a browser security not server security



Cross-Origin Request

A request made from one origin to another origin

Source Origin	Target Origin	
http://localhost:3000	http://localhost:4000	No Access
http://loopkaka.com	https://loopkaka.com	No Access
https://loopkaka.com	https://subdomain.loopkaka.com	No Access
https://google.com	https://loopkaka.com	No Access

CORS (Cross Origin Resource Sharing)

CORS is a mechanism that allows the server to tell the browser which origins are allowed to access its resources.

1. You live in an Apartment - Server
2. Security Guard - Browser
3. Visitor - Another Origin

Frontend (localhost:2000)
 Server(localhost:3000)

(localhost:2000) -----Request-----> Server (localhost:3000) (If yes, process, if No, decline the request)
 Block <----- Not Allowed -----

CORS Headers

1. access-control-allow-origin
2. access-control-allow-method
3. access-control-allow-headers
4. access-control-allow-credentials: true/false
5. access-control-expose-headers

Preflight Request

Before a browser sends a "risky" request (like a POST, PUT or DELETE), It first sends an OPTIONS request to the server.

It's like a polite knock: "Hey, I am coming from Origin A, I want to send a POST request. Is that okay?"

If the server says "Yes" via headers, the browser then sends the real request.

